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Deep learning-driven morphological fingerprinting: rapid, accurate and low-cost pathogen identification via the analysis of dried patterns of droplets

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Abstract

Bloodstream infections (BSIs) are life-threatening conditions with rising mortality, urgently requiring rapid pathogen identification to guide timely antibiotic therapy. Current methods (phenotypic identification, MALDI-TOF MS, molecular techniques) are faced with limitations in cost, turnaround time and accessibility. In this study, it was discovered that the drying of microbial suspensions on slides could lead to species-specific desiccation patterns, which reflects the physicochemical properties and complex interactions of microorganisms during evaporation. Inspired by this interesting phenomenon, an AI-powered platform is proposed for rapid pathogen identification through automated analysis of these patterns. By establishing an image dataset comprising 10,055 desiccation patterns of common BSIs pathogens (*Escherichia coli, Klebsiella pneumoniae, Pseudomonas aeruginosa, Acinetobacter baumannii, Staphylococcus aureus, Enterococcus faecium* and *Candida albicans*), a ResNet-34 deep learning model is trained, achieving a classification accuracy of 91.8% and an area under the receiver operating characteristic curve (AUC-ROC) of 0.99. Following pure culture, identification is completed within 5 min after a simple drying step at 40 °C. With its minimal sample requirement, low cost, and operational simplicity, this platform demonstrates significant potential as a point-of-care diagnostic tool, particularly in resource-constrained regions. This technology offers a promising strategy to combat BSIs and improve patient outcomes.

Keywords Pathogen identification, Droplet evaporation, Desiccation pattern, Deep learning

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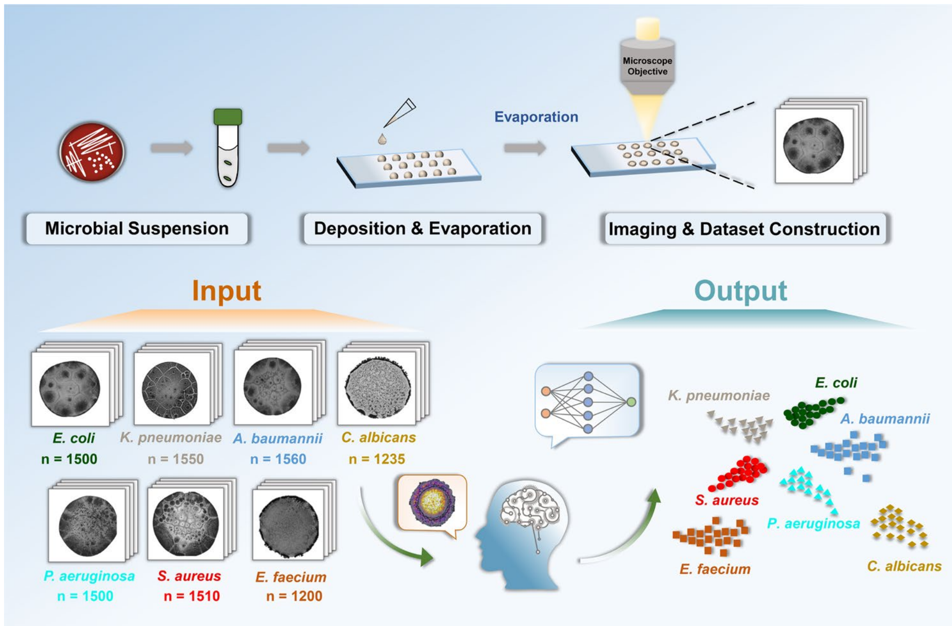
Graphical Abstract



Introduction

Bloodstream infections (BSIs), critical systemic conditions characterized by pathogen invasion and rapid proliferation within the circulatory system, pose a grave threat to global health. It is estimated that more than 250,000 cases of BSIs are recorded in hospitals annually, with mortality rates exceeding 17% (defined as death within 30 days of a positive blood culture), particularly in low-income and resource-limited regions [1–6]. BSIs are caused by diverse pathogens, primarily including Gram-negative bacteria (e.g., *Escherichia coli, Klebsiella pneumoniae, Pseudomonas aeruginosa* and *Acinetobacter baumannii*), Gram-positive bacteria (e.g., *Staphylococcus aureus, Enterococcus faecium*) and fungi (e.g., *Candida albicans*) [1, 2]. Timely and appropriate antibiotic therapy is critical for improving the prognosis of sepsis and reducing mortality. However, the prolonged turnaround time of conventional pathogen culture and identification often leads to empirical broad-spectrum antibiotic administration during emergencies, potentially contributing to the spread of multidrug-resistant pathogens [7, 8]. Studies have demonstrated that rapid and accurate pathogen identification could mitigate antibiotic misuse, aid in determining pathogenicity and guide therapeutic adjustments, collectively facilitating personalized treatment and reducing medical expenses [9–11]. Consequently, the development of a rapid, accurate and simple pathogen identification method holds critical clinical significance for the treatment of BSIs.

In clinical practice, conventional pathogen identification primarily relies on automated systems, typically requiring turnaround time exceeding 8–12 h. The identification of microbial species is based on the detection of metabolic byproduct alterations during the fission of microorganisms, which is time-consuming and labor-intensive [12–14]. In recent years, matrix-assisted laser desorption/ionization time-of-flight mass spectroscopy (MALDI-TOF MS) has emerged as the gold standard for pathogen identification. By analyzing characteristic components like ribosomal proteins to generate species-specific fingerprint spectra, the identification can be completed within several minutes [15–17]. However, the high cost of equipment and dependence on skilled personnel limit its accessibility, particularly in primary care and resource-limited settings. Besides, molecular detection methods such as polymerase chain reaction (PCR) and metagenomic next-generation sequencing (mNGS) also represent emerging approaches for microbial identification [18, 19]. These diagnostic methods also encounter challenges owing to their intricate pre-analytical workflows, equipment costs and time requirements, thereby hindering widespread clinical application. Thus, there is a critical demand for facile, rapid and low-cost diagnostic techniques for the identification of pathogens during the therapy of infections.

Over the past several decades, interest has increasingly focused on the deposition patterns formed by the drying of biologically-relevant sessile droplets [20–22]. The evaporation of biofluids on solid surfaces, such as glass slides, generates complex patterns influenced by the presence and concentration of components (e.g., cells, proteins, metabolites, inorganic ions) and particle interactions/motions. Thus, these patterns could reflect disease-induced compositional changes in biofluids [23–28]. Critically, the desiccation patterns of microbial suspensions also reveal variations in microbial composition and motility [29–31]. For instance, Anusuya Pal et al. observed that surfactants produced by *P. aeruginosa* could induce intense Marangoni flows, resulting in uniform deposition patterns after evaporation. In contrast, *E. coli*, which lacked significant surfactant production, accumulated at the droplet periphery forming ring-shaped deposits [32]. Furthermore, Cheng et al. demonstrated distinct cracking patterns in dried suspensions between wild-type (motile) and mutant (tumbling/non-motile) strains of *E. coli*, probably attributed to the differences in bacterial movement and spatial organization during evaporation [33]. These findings suggest that analyzing the desiccation patterns of biological fluids could offer efficient and straightforward diagnostic strategy for component analysis and disease screening.

In this proof-of-principle application, we have established a facile and accurate platform for the identification of common species in BSIs (*E. coli, K. pneumoniae, P. aeruginosa, A. baumannii, S. aureus, E. faecium* and *C. albicans*) through the interpretation of the dried patterns formed by microbial suspensions on slides. The methodology was based on a strategy of establishing an image dataset of 10,055 dried droplet patterns from seven species. The ResNet-34 deep learning model was then employed to analyze the images and discern the subtle differences [34–36]. The model provides an efficient and objective foundation for dealing with large-scale image datasets, achieving a classification accuracy of 91.8% and an area under the ROC curve (AUC) of 0.99. The entire process requires only 5 min and 2 μL of microbial suspension once pure culture is obtained, eliminating the requirement for sophisticated instruments or specialized personnel. This work demonstrates a user-friendly, low-cost and intelligent tool for timely and precise BSIs treatment.

Materials and methods

Reagents

The blood agar plates (catalogue no. CP0160) used in this study were purchased from Huankai Microbial Technology Co., Ltd. Sodium bicarbonate ($NaHCO_3$, catalogue no. S432108) and 0.9% sodium chloride solution (catalogue no. M485988) were obtained from Aladdin Biochemical Technology Co., Ltd. Erythrocyte lysis buffer was purchased from Shanghai Acmec Biochemical Co., Ltd (catalogue no. AR1010). All reagents were used without further purification. The adhesive glass slides were purchased from Ctiotest Labware Manufacturing Co., Ltd (catalogue no. 158105) and cleaned according to the following steps. The slides were immersed in ethanol and sonicated in an ultrasonic bath for at least 15 min. Then the slides were rinsed in the deionized water (DI) and allowed to be dried in an oven.

Microbial strains

A total of 2011 microbial isolates were used in this study. The composition of the microbiological population was 310 strains of *K. pneumoniae*, 300 strains of *E. coli*, 312 strains of *A. baumannii*, 302 strains of *S. aureus*, 300 strains of *P. aeruginosa*, 240 strains of *E. faecium*, and 247 strains of *C. albicans*. The strains were collected from Nanfang Hospital, Zhujiang Hospital, Guangdong Province, China. All clinical isolates, obtained from urine, blood culture, sputum, and wound specimens, were identified using MALDI-TOF MS according to standard protocols. The study was approved by the Institutional Ethical Committee of Nanfang Hospital (NFEC-202404-Q2).

Dynamic evolution and analysis of the drying of microbial suspensions

A 2 μL aliquot of the microbial suspension was pipetted onto a pre-cleaned glass slide. The slide was immediately transferred to an inverted microscope equipped with 4 × and 40 × objectives for real-time monitoring of the morphological evolution of the microbial suspensions during evaporation at room temperature (30℃). Time-lapse images were captured at 10-s intervals to analyse the dynamic changes in the patterns using an sCMOS camera. Subsequently, the key parameters including the drying kinetics, optical density dynamics and morphological characteristics were further analyzed.

Characterization of desiccation patterns using scanning electron microscopy

The microstructural characterization of the deposits was performed with a Hitachi S-3000N scanning electron microscopy (SEM). The deposits were coated with a 50 nm gold–palladium layer via magnetron sputtering to minimize charging artifacts prior to observation. Secondary electron detection mode was employed for topographic imaging under optimized parameters: 15 kV beam energy and 7.2 mm working distance.

Establishment of image dataset comprising desiccation patterns of microbial suspensions

Microbial strains were cultured on blood agar plates for 16–24 h prior to suspension preparation. Then, several colonies of each strain were picked and used to prepare suspensions in $NaHCO_3$ buffer (100 mM, pH 9.6), adjusting to an optical density of 0.25 at 600 nm.

A two-microliter aliquot of the microbial suspension was deposited onto a pre-cleaned adhesive glass slide, with five replicates per isolate. Then the samples were allowed to be evaporated in the oven under the identical conditions at humidity of 20% ± 3% and temperature of 40 °C ± 0.5 °C for 5 min. After complete evaporation, the desiccation patterns were imaged using a Nikon inverted microscope (ECLIPSE Ti2). Bright-field images were acquired under identical parameters (including light intensity and exposure time) using a 4 × objective. All clinical strains were prepared with the standard procedure to ensure the reproducibility. A dataset consisting of 10,055 images across 7 species (*K. pneumoniae* (n = 1550), *E. coli* (n = 1500), *A. baumannii* (n = 1560), *S. aureus* (n = 1510), *P. aeruginosa* (n = 1500), *E. faecium* (n = 1200), *C. albicans* (n = 1235)) was established for subsequent analysis.

Training and testing of the deep learning model

In order to develop a deep learning system for species classification, the image dataset was split into three independent sets for training, validation, and testing in a 7:1:2 ratio. The proportional distribution of images across different species was maintained in all subsets. The details of these data are presented in Fig. 6.

A uniform image preprocessing pipeline was applied to all images in the training, validation, and testing sets. All images were resized to 224 × 224 pixels, and pixel values were normalized to [0, 1]. The scaled values were standardized using the mean and standard deviation derived from the ImageNet dataset. The original images were acquired under highly standardized conditions (including light intensity and exposure time) with minimal noise and uniform illumination. Consequently, no additional preprocessing techniques were employed.

The ResNet-34 architecture was employed for species classification based on desiccation patterns. The model was initialized with pre-trained weights from ImageNet. The final fully-connected layer was replaced and fine-tuned on the target dataset during transfer learning. The network primarily consists of: (1) an initial feature extraction module, (2) four residual stages incorporating identity mapping, and (3) a classification head. The model was optimized using stochastic gradient descent (SGD) with a learning rate of 0.0002 and a momentum of 0.9. Detailed architectures are provided in the Supplementary Information.

The accuracy, precision, recall, specificity, F1-score and AUC were analyzed to evaluate the performance and effectiveness of the model (True Positive: TP, True Negative: TN, False Positive: FP, False Negative: FN).

$$
\begin{gathered}
Accuracy = (TP + TN) / (TP + FP + TN + FN) \\
Precision = TP / (TP + FP) \\
Recall = TP / (TP + FN) \\
Specificity = TN / (TN + FP) \\
F1\_Score = 2 * Precision * Recall / (Precision + Recall)
\end{gathered}
$$

The ROC curve graphically represents the performance of classification model across all classification thresholds. It plots the True Positive Rate (TPR) against the False Positive Rate (FPR), where TPR = TP/(TP + FN) and FPR = FP/(FP + TN). The AUC is calculated by integrating the TPR with respect to the FPR over the interval [0, 1]. It represents the probability that the model ranks a random positive instance higher than a random negative instance.

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(\mathrm{FPR})\mathrm{d}(\mathrm{FPR})$$

Quantitative analysis of the characteristics of desiccation patterns

The images were analyzed using RS Image software. All images were adjusted to 8-bit gray values. An appropriate threshold value was set to segment features from the background. For the analysis of the total area and mean gray value of desiccation patterns, the threshold was set according to the "MaxEntropy" algorithm provided by the software to automatically identify the features of the patterns and output the analysis results. The morphology and grayscale changes of droplets during the evaporation process were further analyzed.

Collection of microbial cells from positive blood cultures

A five-milliliter aliquot from positive blood culture specimen was centrifuged at 1,000 × g for 5 min, and the supernatant was collected. The supernatant was then subjected to high-speed centrifugation at 15,600 × g for 3 min. The resulting pellet was incubated with erythrocyte lysis buffer for 8 min and subsequently harvested by centrifugation (15,600 × g, 3 min). After two washes with pure water under identical centrifugal conditions, the final pellet was resuspended in 100 mM $NaHCO_3$ buffer. The optical density of the suspension was measured at 600 nm and adjusted to the target concentration using the same buffer. The suspensions acquired were then dried under the controlled conditions for the analysis of desiccation patterns.

Results

Species-specific morphological fingerprints in the desiccation patterns from microbial suspensions

To investigate the characteristics of desiccation patterns formed by microbial suspensions, we selected seven common bloodstream infection pathogens—*K. pneumoniae*, *E. coli*, *A. baumannii*, *P. aeruginosa*, *S. aureus*, *E. faecium*, and *C. albicans* as model strains based on epidemiological data [1, 2]. Pure colonies were picked and resuspended in carbonate buffer to prepare microbial suspensions. Carbonate buffer, a kosmotropic agent in the Hofmeister series, was selected to amplify protein–protein interactions through preferential hydration effects, thereby enhancing pattern formation via salting-out mechanisms. Following the procedures detailed in the Methods section, deposition experiments were performed by dispensing 2 μL aliquots onto glass substrates and the desiccation patterns were captured with consistent illumination parameters. This standardized process ensures that the differences in desiccation patterns primarily originate from the inherent biological properties of the microorganisms.

Figure 1 illustrates the typical characteristics of the desiccation patterns across seven species. The droplet depositions were reproducible and exhibited an average diameter of 2.0 ± 0.1 mm (Fig. 1c). Regarding the structural features, patterns derived from the same species demonstrated a high degree of similarity, while distinct differences were observed between different species. As depicted in Fig. 1b, dried droplets of *K. pneumoniae* exhibited a uniform deposition accompanied by prominent cracking structures. Conversely, dried droplets from *E. coli, A. baumannii, P. aeruginosa, S. aureus, and E. faecium* formed punctate structures with varying sizes and quantities, surrounded by minor cracking. Through the threshold analysis, notable differences were identified in terms of the area proportion of the region of interest (ROI) as well as the quantity of punctate structures within the dried droplets across different species (Fig. 1d). As for *C. albicans*, the morphology of dried droplets primarily appeared as a filamentous network structure that showed significant divergence compared to the aforementioned patterns (Fig. 1b). Additionally, deposits formed from the same sample exhibited excellent reproducibility in droplet size and grayscale characteristics, indicating high stability and repeatability throughout the process (Figure S1). Furthermore, the desiccation patterns of microbial suspensions appeared consistently at heating temperatures ranging from 30 °C to 40 °C during evaporation (Figure S2), confirming the robustness of the method and its applicability for on-site detection under varying environmental conditions. These findings collectively suggested that species-specific morphological features are present in the desiccation patterns across different species, potentially associated with variations in microbial cellular composition and movement patterns during droplet evaporation. Further analysis of the fingerprints of these dried deposits could potentially be developed into a novel, rapid and facile approach for species identification.

Dynamic evolution of the drying of microbial droplets

To further explore the fundamental mechanisms governing the formation of desiccation patterns, we investigated the morphological dynamics during the evaporation of microbial droplets. The microbial suspension was deposited onto a glass slide and immediately transferred to an inverted microscope equipped with 4 × and 40 × objectives for real-time monitoring at room temperature (30 °C). Imaging with a 40 × objective enables the visualization of higher-resolution details of crystal structures, microbial arrangements, and dynamic processes during evaporation (e.g., coffee-ring formation, crystal growth). Evaporation at 30 °C extended the drying time from approximately 1020 s to 1100 s without significantly altering the resulting desiccation patterns, as shown in Figure S2. Figure 2 depicts the drying evolution of droplets. This evaporation process exhibited a two-stage progression: (I) moisture vaporization stage and (II) crystallization stage. Throughout stage I, the droplets presented a uniform appearance, and their morphological characteristics and evaporation dynamics remained consistent across all species. Divergence occurred only during the crystallization stage, when crystal structures emerged from the supersaturated solution.

(I) Moisture vaporization stage: Upon deposition on the glass slide surface, the sessile droplet initially exhibited a spherical cap morphology. During evaporation (approximately 0 to 1000 s), the droplet gradually flattened. Concurrently, dynamic imaging revealed a continuous decrease in the gray value (Fig. 2, Fig. 3, Movies S1-S7). All images for grayscale analysis were acquired with a 4 × objective, encompassing the entire droplet. Simultaneously, microscopic observations showed that individual microorganisms continuously migrated from the central region of the droplet towards the edge, where they accumulated and formed a ring-like structure (Fig. 4). This phenomenon is known as the coffee-ring effect (Figure S3). At the beginning of evaporation, due to the higher evaporation rate at the edge of the droplet, a radially outward capillary flow from center to the periphery was developed [37, 38]. This flow transported suspended particles such as microorganisms to the periphery, leading to the enrichment and ring formation.

(II) Crystallization stage: During the latter stage of droplet evaporation, the gradual reduction in water content led to an increase in $NaHCO_3$ concentration. When the concentration exceeded its solubility, the solution became supersaturated, initiating the crystallization of $NaHCO_3$. As illustrated in Fig. 4, Movies S8-S14, $NaHCO_3$ nucleation occurred at the periphery of the droplet and rapidly propagated towards the central region, forming a skeletal crystalline structure within 20–50 s. This crystallization event was marked by a distinct, rapid increase in the gray value. The quantitative

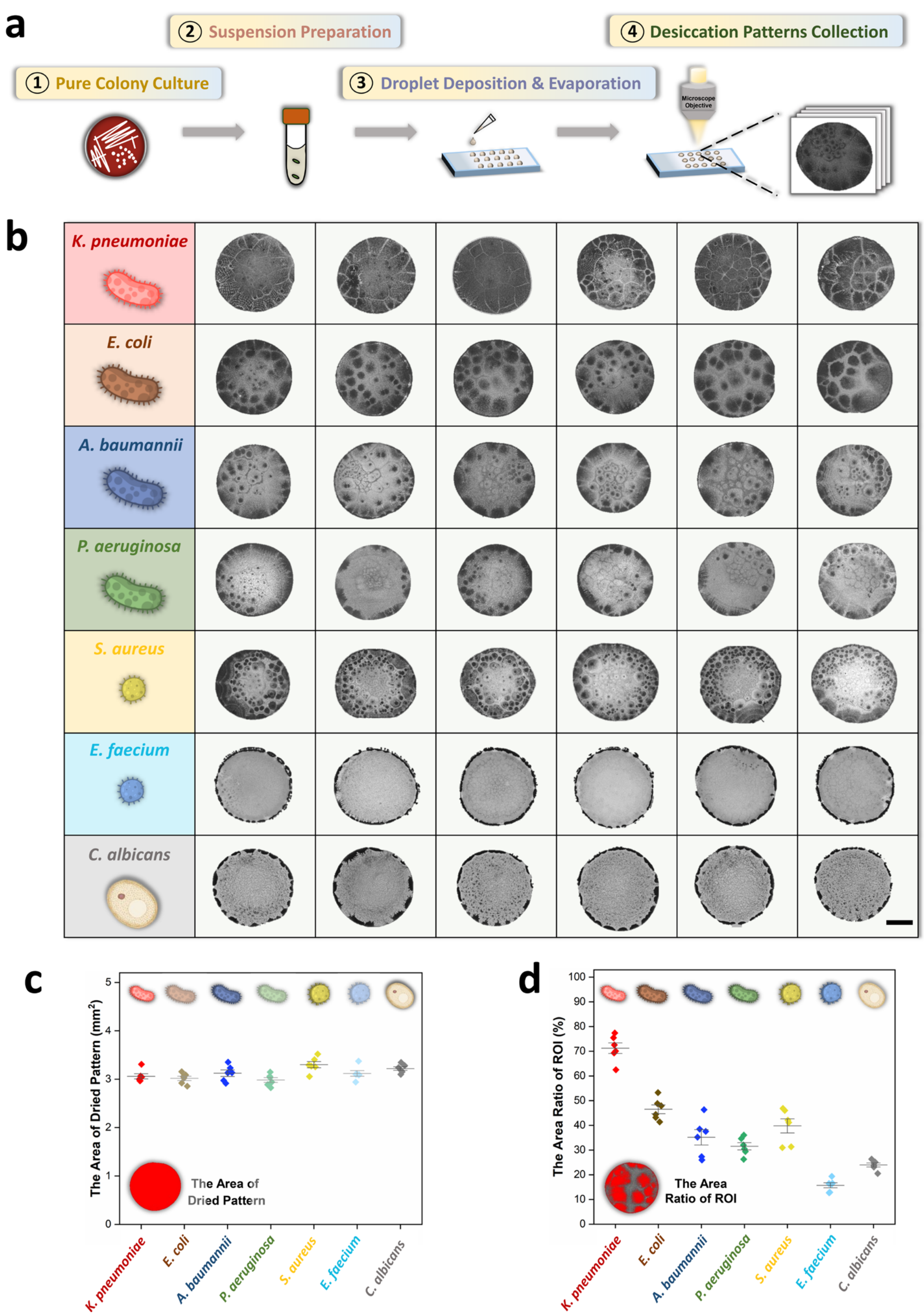


Fig. 1 Species-specific desiccation patterns of microbial suspensions. (**a**) The workflow for the deposition and evaporation of microbial suspensions. (**b**) Distinct morphological characteristics of dried microbial suspensions across seven species (*K. pneumoniae, E. coli, A. baumannii, P. aeruginosa, S. aureus, E. faecium* and *C. albicans*). Scale bar: 500 µm. (**c**, **d**) Quantitative analysis of the total pattern area and the area ratio of the region of interest (ROI)

grayscale analysis provided a clear signature of the crystallization phase, which was consistent across species but exhibited subtle, species-specific variations in timing and intensity (Fig. 3).

During this crystallization process, individual microorganisms were mechanically entrapped within the interstitial spaces of the growing crystals, thereby suppressing their migration toward the edge of the droplet. Additionally, electrostatic repulsion between the negatively charged crystal surfaces ($NaHCO_3$) and microbial cells (typically carrying negative surface charges) likely confined the microorganisms to specific regions. After complete evaporation, crystal-microorganism composite structures were observed at the periphery of the droplet,

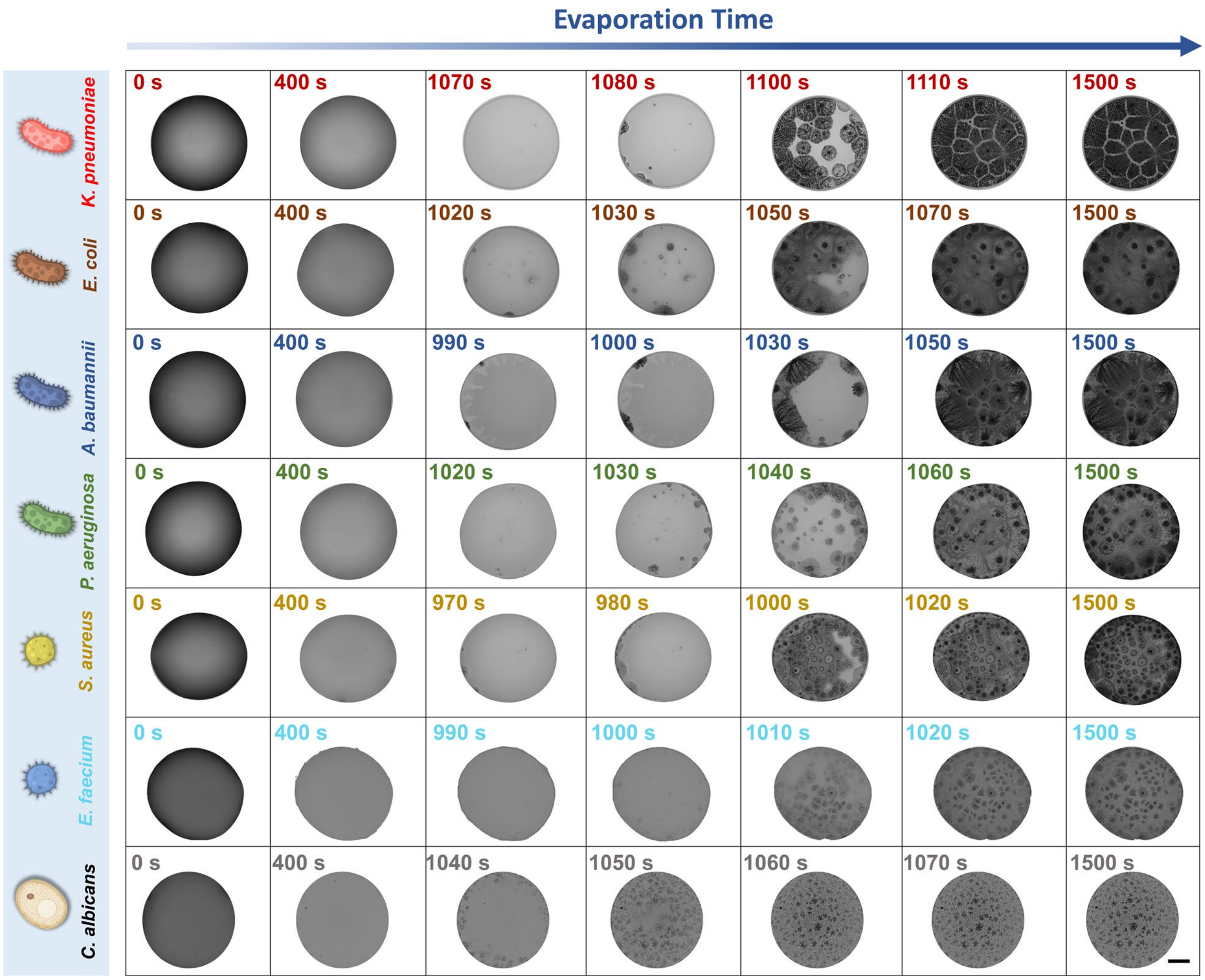


Fig. 2 The dynamic evolution of microbial droplets dried at 30 °C was recorded using a 4 × objective. Scale bar: 500 µm

while the central region primarily exhibited aggregates of microbial cells alongside sparse composite deposits. Collectively, the species-specific variations in crystallization dynamics revealed by quantitative grayscale analysis (Fig. 3) likely stem from differential spatial distribution and entrapment of microbial cells within the crystal matrix, ultimately contributing to the formation of distinct, species-specific desiccation patterns.

Notably, the deposition patterns of microbial suspensions after evaporation exhibited high intra-species consistency but significant inter-species variation. To further explore this phenomenon, SEM was employed to analyze the microscopic composition. As shown in Fig. 5, the peripheral region predominantly consisted of crystal-microorganism composites, whereas the central region comprised distinct microbial deposition aggregates. Evidence from droplet drying studies indicates that the morphology of desiccation patterns is fundamentally governed by the physicochemical properties of the solutes [39]. In line with this, the observations by SEM (Fig. 5) revealed distinct microstructural arrangements in the central and peripheral regions of dried droplets, which exhibited species-specificity. We therefore hypothesize that these pattern differences arise from interspecies divergence in key biophysical properties—such as membrane flexibility, surface roughness, surface charge, and motility [29–33, 40–43]. These inherent characteristics might modulate microbial interactions with crystallizing solutes and their responses to evaporation-driven hydrodynamic forces. Given the multifactorial nature of pattern formation, future systematic investigations are essential to fully elucidate the mechanistic principles underlying pattern formation.

Establishment of image dataset consisting of the desiccation patterns

To explore the potential and feasibility of utilizing the desiccation patterns of microbial suspension as a rapid, cost-effective tool for pathogen identification, we collected a total of 2,011 clinical isolates across seven species

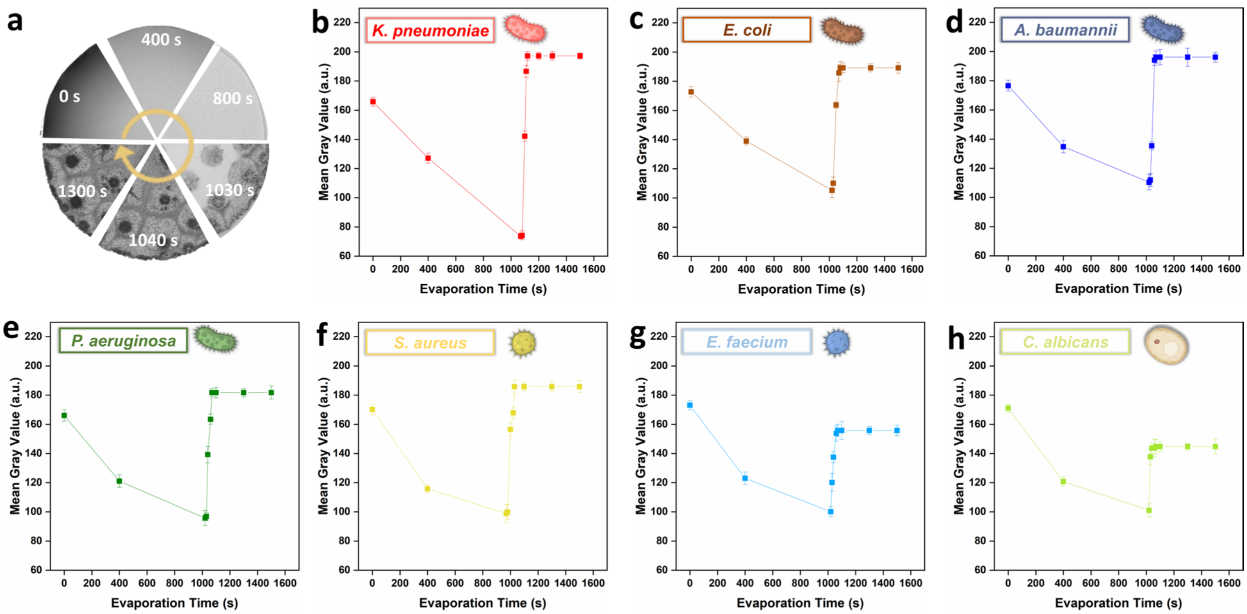


Fig. 3 Analysis of the mean gray value of patterns during evaporation across different species. (**a**) Schematic diagram depicting the time-lapse grayscale changes of a droplet during the drying process. (**b**, **c**, **d**, **e**, **f**, **g**, **h**) Dynamic grayscale evolution during droplet evaporation for different species (*K. pneumoniae, E. coli, A. baumannii, P. aeruginosa, S. aureus, E. faecium and C. albicans*). Error bars represent standard deviation from three replicates

to construct an image dataset (the number of strains per species is detailed in the Methods section). A deep learning-based prediction model would be employed for species classification in subsequent assays, addressing the subjectivity and variability from the manual interpretation as well as discovering the subtle associations which are undetectable and unexplainable by human observers (Scheme 1).

As described previously, microbial suspensions were prepared in 100 mM carbonate buffer following pure colony culture. The initial concentration of microbial suspensions critically determined the characteristics of desiccation patterns. Specifically, a high concentration (1 OD_{600}) promoted cell aggregation into homogeneous deposits (e.g., *K. pneumoniae*), while a low concentration (0.125 OD_{600}) resulted in sparse cellular distributions that lacked identifiable fingerprint-like signatures (Figure S4, Figure S5). To ensure optimal reproducibility, the initial concentration of all microbial suspensions were standardized to 0.25 OD_{600} for all subsequent assays. For each isolate, five replicates of a 2-μL suspension were deposited onto a glass slide and allowed to evaporate at 40℃. The desiccation patterns were then captured using a 4 × objective for image acquisition, as described in the Methods section. Thus, an image dataset comprising 10,055 images of desiccation patterns from microbial suspensions across seven species was established. The images were further labeled according to the source of species (*K. pneumoniae* “0”, *E. coli* “1”, *A. baumannii* “2”, *S. aureus* “3”, *P. aeruginosa* “4”, *E. faecium* “5”, *C. albicans* “6”). To develop a deep learning system for species classification, the image dataset was randomly partitioned into three independent subsets for training, validation and testing at a ratio of 7:1:2. Details of the dataset are presented in Fig. 6.

Deep learning-driven species identification based on desiccation patterns

To validate the feasibility of deep learning driven single-species identification via desiccation pattern analysis, we employed the ResNet-34 architecture for the subsequent analysis. It was selected for its optimal balance between depth and representational capacity, as its residual connections mitigate the vanishing gradient problem and facilitate learning fine-grained features, which are essential for discerning subtle inter-species differences. Preliminary benchmarking within the ResNet series corroborated these technical advantages, confirming that ResNet-34 outperformed ResNet-18 (Figure S6, Table S1). To circumvent the typical demands of large annotated datasets and extensive training time, we utilized transfer learning by fine-tuning a ResNet-34 model pre-trained on ImageNet [44]. The fine-tuned model was evaluated on a testing set and selected as the final predictive model.

The architecture and training details of the algorithm are provided in the Supplementary Information and summarized in Fig. 7. ResNet-34 consists of three principal components: (1) an initial convolution-pooling module that captures low-level spatial patterns from input images and reduces feature map dimensions while retaining critical information; (2) residual stages with skip connections,

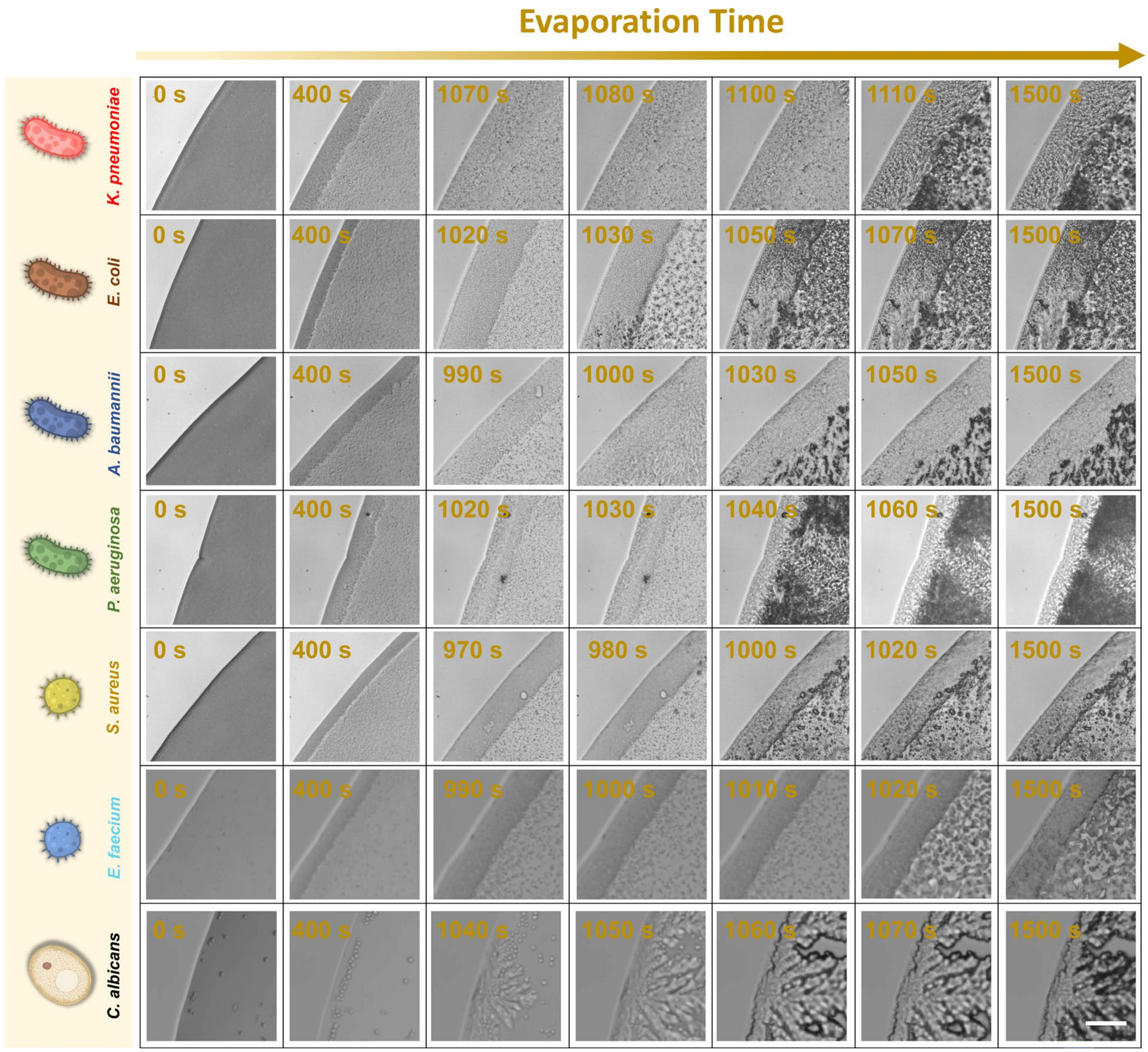


Fig. 4 The dynamic evolution of microbial droplets dried at 30 °C was recorded under a microscope (40 × objective). Scale bar: 20 μm

which form the core innovation of ResNet-34, enabling effective training of deep networks by mitigating gradient vanishing and stabilizing learning dynamics, and (3) a classification module that employs global average pooling to condense spatial features, followed by a fully connected layer and softmax activation to generate probabilistic classification outputs [45]. Subsequently, the AUC, accuracy, precision, recall, specificity and F1_Score were analyzed to evaluate the performance of the model (Fig. 8, Figure S7). The model demonstrated exceptional discriminatory power in the classification task, achieving a near-perfect AUC of 0.99 and a robust accuracy of 91.8%. The precision, recall, specificity and F1-Score reached 0.9021, 0.8992, 0.9225 and 0.8993 respectively. The high precision and specificity indicate a strong ability to minimize false positives, while the high recall reflects the effective minimization of false negatives. The balanced F1-score further confirms the model's robustness across all microbial categories. Furthermore, the AUC provides a robust, threshold-independent evaluation metric that mitigates potential biases from class imbalance. This is particularly valuable as skewed class distribution can inflate accuracy and render threshold-sensitive metrics like precision and recall less reliable. In contrast, by assessing performance across all classification thresholds, AUC offers a more comprehensive and stable measure of the model's inherent discriminative ability. Collectively, these results underscore the model's strong potential for high-confidence species identification.

Detailed species-specific model performance is presented in Fig. 8b-d. Overall, the AUC for each species ranged from 0.96 to 1.00 (Fig. 8b), with slightly lower

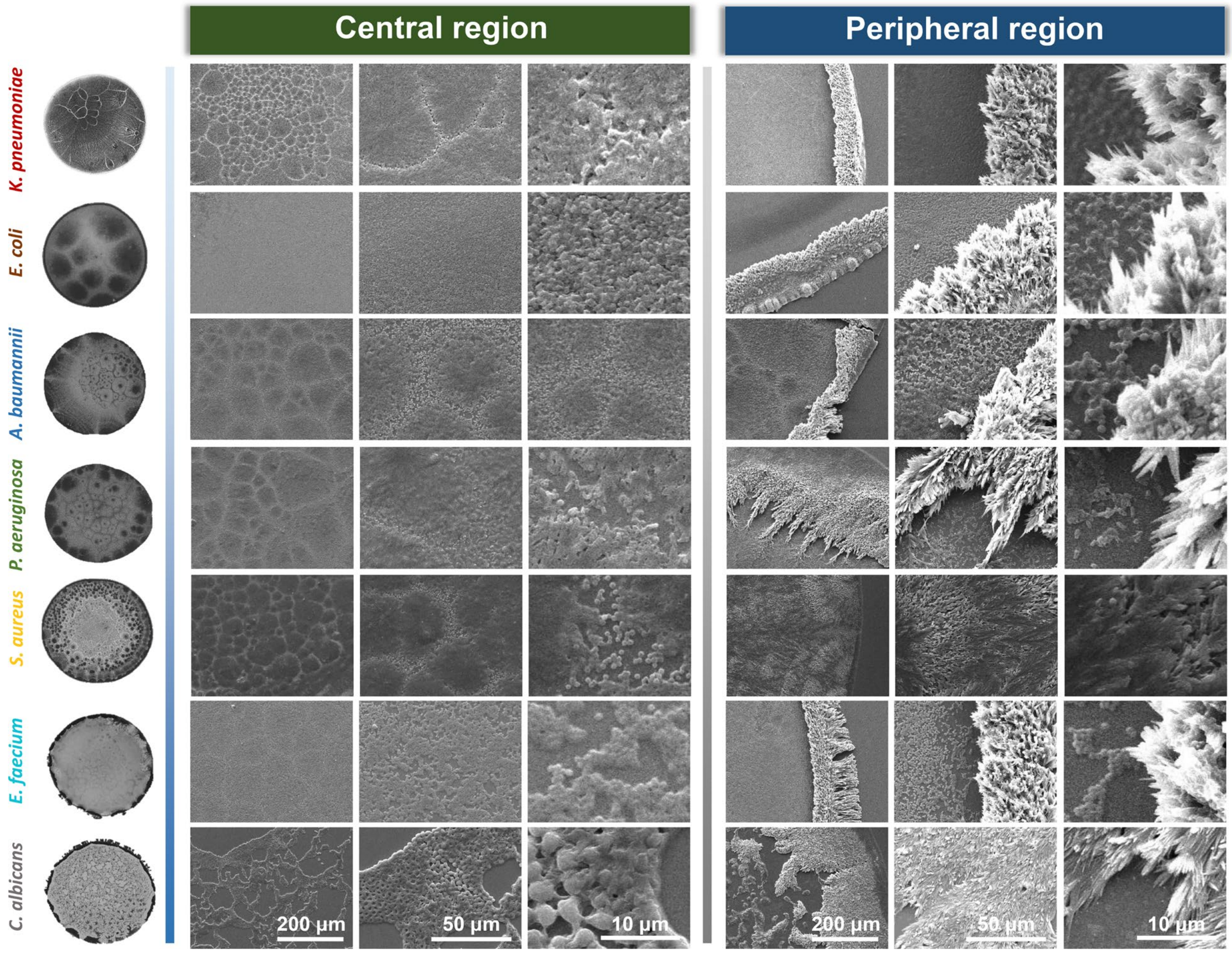
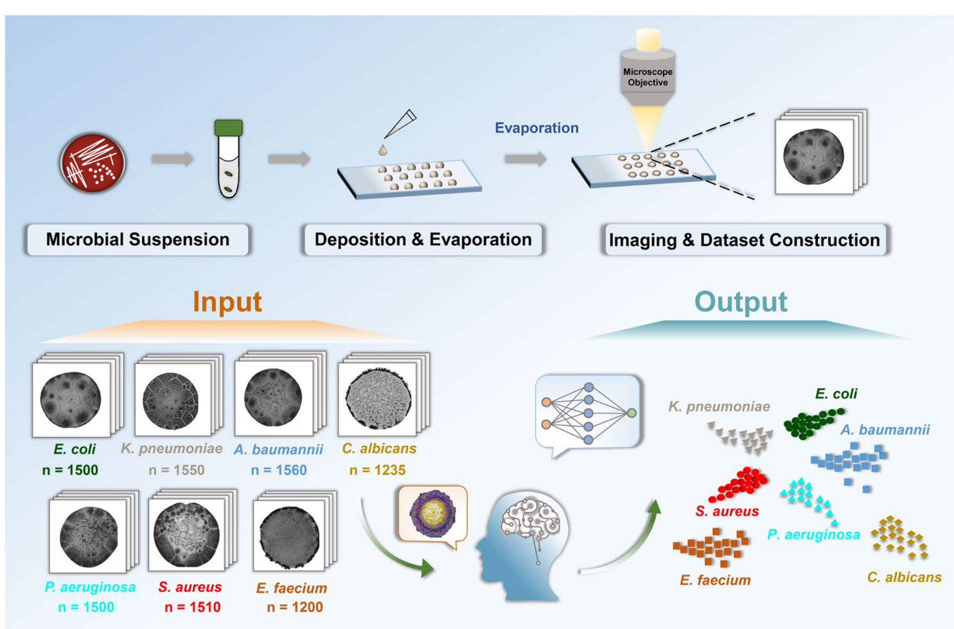


Fig. 5 Microscopic analysis of the central and peripheral regions in desiccation patterns across different species was performed using SEM



Scheme 1. Schematic of the deep learning-driven assay protocol for species classification using desiccation patterns of microbial suspension

Strains	Training Set	Validation Set	Testing Set
K. pneumoniae	1085	155	310
E. coli	1050	150	300
A. baumannii	1090	160	310
S. aureus	1050	150	310
P. aeruginosa	1050	150	300
E. faecium	840	120	240
C. albicans	860	125	250
Total Images	7025	1010	2020

Fig. 6 Statistical information of the image dataset formed by the desiccation patterns of microbial suspensions across seven species

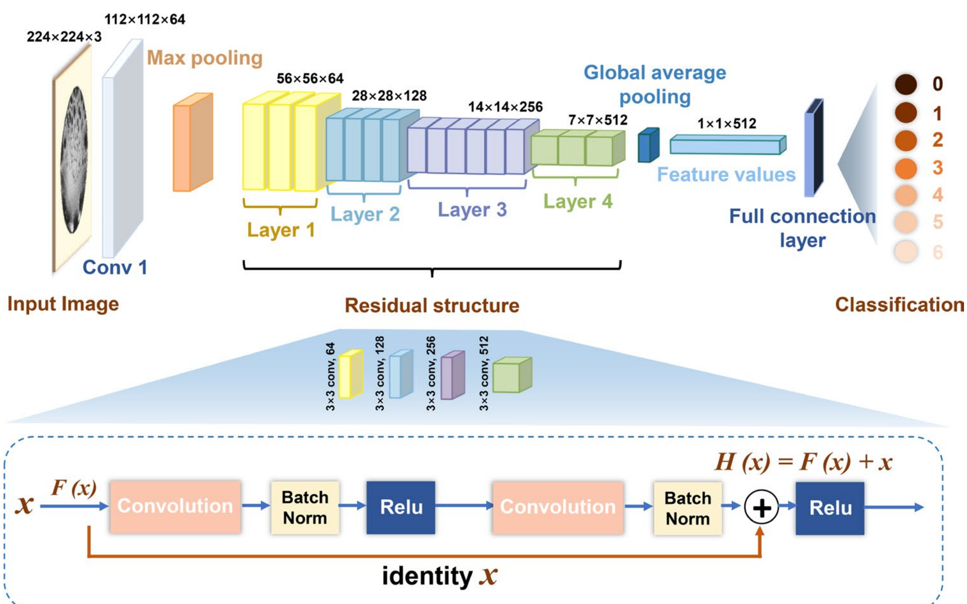
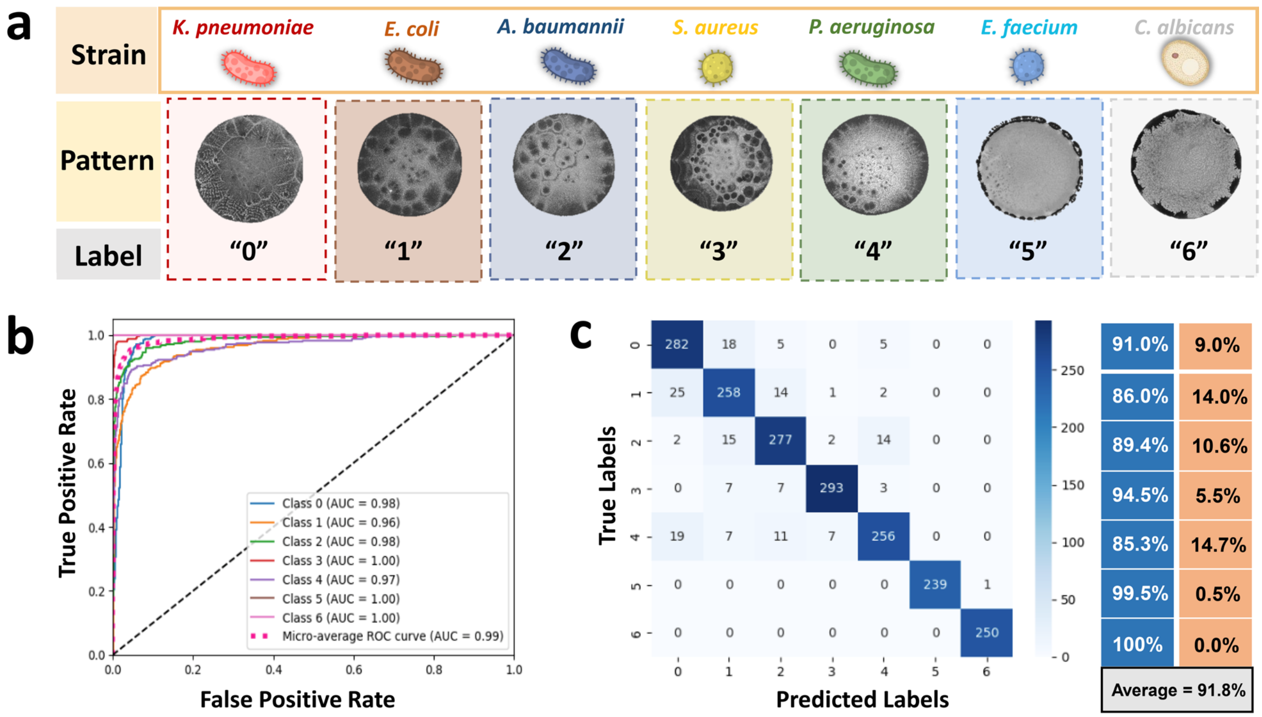


Fig. 7 The major architecture of the ResNet 34 model

accuracy for *E. coli* and *P. aeruginosa* (Fig. 8c). This minor reduction in accuracy might arise from intra-species morphological and phenotypic diversity (e.g., mucoid *vs.* non-mucoid phenotypes), which could introduce variability in desiccation patterns and consequently impact model performance. To investigate such phenotypic diversity, representative non-mucoid and mucoid strains of *E. coli* and *P. aeruginosa* were selected for comparative analysis. As shown in Figure S8, non-mucoid and mucoid strains of *E. coli* and *P. aeruginosa* exhibited differences in colony morphologies and desiccation patterns. The desiccation patterns of mucoid strains exhibited a reduction in the central punctate structures compared to those of non-mucoid strains. Accordingly, the model misclassified one of the mucoid *E. coli* strains as *P. aeruginosa*. These findings highlight the necessity of incorporating strain-level phenotypic diversity into future training datasets to improve model generalizability and clinical utility. Additionally, we retrospectively evaluated the effect of antimicrobial susceptibility variability on model performance by



d

Label	Specificity	Precision	Recall	F1_Score	AUC
K. pneumoniae “0”	0.9635	0.8598	0.9097	0.8840	0.98
E. coli “1”	0.9560	0.8459	0.8600	0.8529	0.96
A. baumannii “2”	0.9653	0.8822	0.8935	0.8935	0.98
S. aureus “3”	0.9866	0.9679	0.9452	0.9564	1.00
P. aeruginosa “4”	0.9663	0.9143	0.8533	0.8827	0.97
E. faecium “5”	0.9995	1.0000	0.9958	0.9979	1.00
C. albicans “6”	0.9995	0.9960	1.0000	0.9980	1.00
Model	0.9225	0.9021	0.8992	0.8993	0.99

Fig. 8 Performance evaluation of the ResNet-34 model for species classification. (**a**) Distribution of image labels across species. (**b**, **c**) ROC curves and the corresponding confusion matrix of the seven-class classifier. (**d**) Model performance metrics including precision, recall, specificity, F1-score and AUC

analyzing critical resistance profiles among the strains in the testing set, such as carbapenem resistance in *Enterobacter* and oxacillin resistance in *S. aureus*. Classification accuracy remained comparable between susceptible and resistant subgroups for *K. pneumoniae* (92.7% vs. 88.7%), *E. coli* (85.5% vs. 87.9%) *and S. aureus* (94.6% vs. 94.3%) (Figure S9, Table S2). These results suggest that species-specific morphological fingerprints are resilient to these resistance mechanisms. Future work will be necessary to determine whether this finding extends to a broader spectrum of resistance mechanisms.

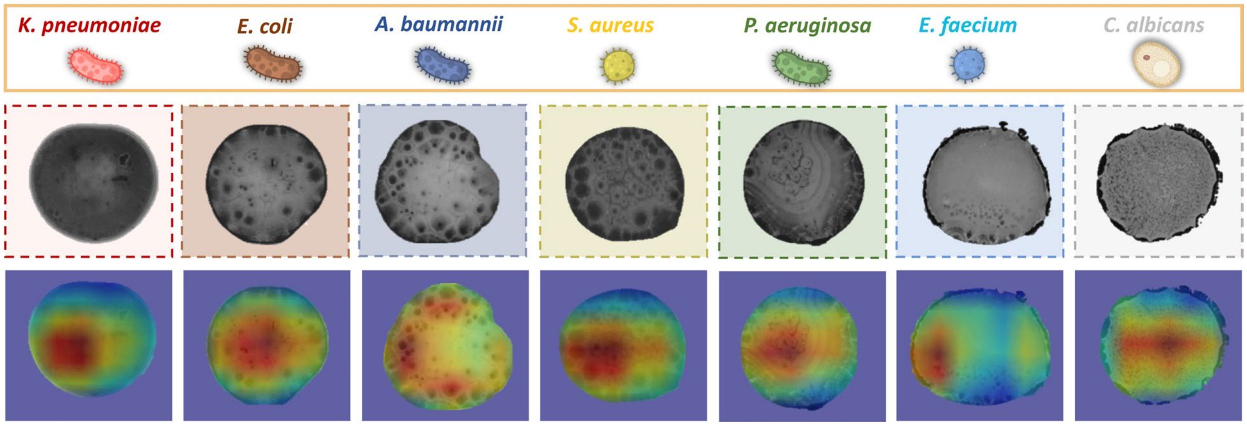


Fig. 9 The Grad-CAM visualizations of the desiccation patterns from microbial droplets across different species

To explore whether the approach could be extended to direct blood or clinical fluid samples, we enriched microbial cells from 24 positive blood culture samples (including *K. pneumoniae*, *E. coli*, and *S. aureus*) using gradient centrifugation and cell lysis to remove erythrocytes and leukocytes. The pellets were resuspended in 100 mM $NaHCO_3$ buffer and the desiccation patterns of the microbial suspensions for each sample were acquired and analyzed following the standardized method described above. The prediction results demonstrated the preliminary feasibility of the approach, except for one of the *E. coli* isolates that was misclassified as *A. baumannii* (Figure S10). However, challenges remain for direct application to clinical specimens due to sample complexity, such as variable microbial loads and interference from host components. Future improvements could integrate short-term enrichment or microfluidic-based concentration to enhance microbial recovery without relying on prolonged culture. In summary, these results demonstrated that species-specific desiccation patterns of microbial suspensions could be efficiently and accurately classified using deep learning algorithms, highlighting the potential for establishing a facile, rapid and automatic platform for the identification of pathogens.

Grad-CAM analysis revealed dominant feature in the prediction model

To elucidate the decision-making mechanisms of the deep learning model in image interpretation, gradient-weighted class activation mapping (Grad-CAM) was employed. This interpretability technique computes the gradients of the class scores with respect to the activations in the final convolutional layer. The resulting gradient magnitudes reflect the importance of each neuron in the feature map for the classification decision, thereby highlighting the discriminative image features used by the model to differentiate among classes.

Figure 9 demonstrates both the desiccation patterns of microbial droplets and their corresponding Grad-CAM visualizations. Critical regions with elevated gradient magnitudes (indicated by warm colors in the heatmap) represent regions that predominantly influenced the classification outcomes. Our findings revealed that the central regions of desiccation patterns, characterized by high activation weights, contributed significantly to the discrimination while the peripheral background elements provided negligible predictive information. The results of Grad-CAM analysis were consistent with those obtained from SEM, indicating that differences in microbial arrangement and crystal structure in the central region of desiccation patterns were dominant factors influencing the decision-making mechanism of the prediction model.

Discussion

In this study, we present a simple, rapid, and cost-effective diagnostic platform for species identification by leveraging the deep learning-driven analysis of desiccation patterns from microbial suspensions. First, we demonstrated that the desiccation patterns from microbial suspensions encoded species-specific morphological signatures, driven by complex interactions between microorganisms, hydrodynamic forces, and solute crystallization during droplet evaporation [21, 25]. Second, we constructed an image dataset comprising 10,055 desiccation patterns across seven common pathogens. Subsequently, we developed an AI-driven platform that achieved a species-level identification accuracy of 91.8% with an AUC of 0.99. This method offers significant advantages including low cost, operational simplicity, and high accuracy.

In contrast, established gold-standard methods such as MALDI-TOF MS, PCR, or mNGS excel in sensitivity and broad-spectrum identification [46–48], but they often entail long turnaround times, high costs, and centralized laboratory settings. In addition, existing imaging or AI-based microbial identification techniques including morphology-based microscopic detection and Raman/Surface Enhanced Raman Spectroscopy (SERS) coupled with machine learning typically rely on high-resolution cellular data, fluorescent labeling, and sophisticated instruments (e.g., super-resolution or Raman microscopes) [49–54], which might limit their accessibility in resource-limited settings. In comparison, the novel approach requires no staining, uses minimal sample volume, and needs only basic bright-field microscopy, significantly lowering both cost and operational complexity. By eliminating the reliance on sophisticated instrumentation and specialized personnel, this method is particularly suitable for minimally trained users in resource-limited environments.

The current study primarily focuses on demonstrating the feasibility of deep learning-driven desiccation pattern analysis to identify single-species infections, which serves as a foundational step. A key future direction is the systematic expansion of the pathogen library to include a broader spectrum of clinically relevant species, particularly rare and emerging pathogens. This effort will improve the model's generalizability by training it on more diverse morphological signatures. Furthermore, given that polymicrobial infections are common and diagnostically challenging [55, 56], adapting this platform for such cases is a priority. Promising strategies include developing multi-label classification models trained on synthetic mixed suspensions, or using spatial pattern segmentation to resolve overlapping features. These enhancements would significantly boost the clinical utility of novel approach in complex, real-world scenarios.

Furthermore, a deeper investigation into the mechanisms underlying the formation of desiccation patterns from microbial suspensions is required, focusing on key physicochemical processes such as capillary flow, the coffee-ring effect, and crystallization dynamics [21, 25]. Concurrently, relevant biophysical parameters of the microorganisms such as surface hydrophobicity, membrane elasticity, secretory metabolites, and other intrinsic biological traits could be systematically investigated to clarify their regulatory roles in pattern formation [40, 41]. Integrating these mechanistic insights with quantifiable biological features would enable the construction of a multimodal data architecture. Such efforts are expected not only to enhance classification performance but also to provide a more explainable framework for AI-driven diagnosis, thereby establishing a more robust and interpretable connection between biophysical mechanisms and algorithmic decision-making.

Although limitations persist (e.g., prerequisite pathogen isolation/culture and a limited number of distinguishable species), future improvements in sample preparation, database scale and diversity, and algorithmic performance are poised to broaden its applicability. Building upon this foundation, a key future direction involves integrating ubiquitous mobile technology. A dedicated smartphone application, coupled with an affordable clip-on lens, could enable the capture of desiccation patterns directly at the point of care. These images would be transmitted to a cloud server running the optimized AI model for analysis and species identification, with results swiftly returned to the user's device. This mobile-to-cloud architecture promises to democratize access to advanced diagnostics, transforming smartphones into portable identification tools.

Conclusion

Rapid and accurate pathogen identification is essential for the treatment of BSIs. In this work, we revealed that the desiccation patterns of microbial suspensions encode species-specific morphological signatures. By leveraging deep learning-driven analysis, we established a user-friendly, low-cost, and intelligent platform for species classification, providing a promising strategy for pathogen identification. This platform shows the potential to be integrated with mobile technology for point-of-care applications, particularly in settings where traditional laboratory infrastructure is scarce.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12951-025-03923-9.

Additional file 1
Additional file 2
Additional file 3
Additional file 4
Additional file 5
Additional file 6
Additional file 7
Additional file 8
Additional file 9
Additional file 10
Additional file 11
Additional file 12
Additional file 13
Additional file 14
Additional file 15

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Author contributions

Shujuan Guan, Ruyue Yang, Danning Deng and Huilin Long contributed equally to this work. Conceptualization: Guan SJ, Yang RY, Hu XM, Chen DQ. Method: Guan SJ, Yang RY, Deng DN, Long HL. Investigation: Guan SJ, OU ZH, Ge XX, Jiang XJ. Project management: Guan SJ, Hu XM, Chen DQ. Supervision: Hu XM, Chen DQ. Writing First Draft: Guan SJ, Yang RY, Deng DN, Long HL. Writing—Reviewing and Editing: Guan SJ, Hu XM, Chen DQ.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was approved by the institutional ethical committees of the Nanfang Hospital (NFEC-202404-Q2).

Consent for publication

All authors have approved the manuscript and agree for the submission.

Competing interests

The authors declare no competing interests.

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References

1. Holmes CL, Albin OR, Mobley H, Bachman MA. Bloodstream infections: mechanisms of pathogenesis and opportunities for intervention. Nat Rev Microbiol. 2025;23:210–24.
2. Kontula K, Skogberg K, Ollgren J, Jarvinen A, Lyytikainen O. Population-based study of bloodstream infection incidence and mortality rates, Finland, 2004–2018. Emerg Infect Dis. 2021;27:2560–9.
3. Diekema DJ, Hsueh PR, Mendes RE, Pfaller MA, Rolston KV, Sader HS, et al. The microbiology of bloodstream infection: 20-year trends from the SENTRY antimicrobial surveillance program. Antimicrob Agents Chemother. 2019;63:e00355-19.
4. Wisplinghoff H, Bischoff T, Tallent SM, Seifert H, Wenzel RP, Edmond MB. Nosocomial bloodstream infections in US hospitals: analysis of 24,179 cases from a prospective nationwide surveillance study. Clin Infect Dis. 2004;39:309–17.
5. Marra AR, Camargo LF, Pignatari AC, Sukiennik T, Behar PR, Medeiros EA, et al. Nosocomial bloodstream infections in Brazilian hospitals: analysis of 2,563 cases from a prospective nationwide surveillance study. J Clin Microbiol. 2011;49:1866–71.
6. Verway M, Brown KA, Marchand-Austin A, Diong C, Lee S, Langford B, et al. Prevalence and mortality associated with bloodstream organisms: a population-wide retrospective cohort study. J Clin Microbiol. 2022;60:e242921.
7. Micek ST, Hampton N, Kollef M. Risk factors and outcomes for ineffective empiric treatment of sepsis caused by Gram-negative pathogens: stratification by onset of infection. Antimicrob Agents Chemother. 2018;62:e00007-18.
8. Marquet K, Liesenborgs A, Bergs J, Vleugels A, Claes N. Incidence and outcome of inappropriate in-hospital empiric antibiotics for severe infection: a systematic review and meta-analysis. Crit Care. 2015;19:63.
9. Yahav D, Franceschini E, Koppel F, Turjeman A, Babich T, Bitterman R, et al. Seven versus 14 days of antibiotic therapy for uncomplicated Gram-negative bacteremia: a noninferiority randomized controlled trial. Clin Infect Dis. 2019;69:1091–8.
10. Holland TL, Raad I, Boucher HW, Anderson DJ, Cosgrove SE, Aycock PS, et al. Effect of algorithm-based therapy vs usual care on clinical success and serious adverse events in patients with staphylococcal bacteremia: a randomized clinical trial. JAMA. 2018;320:1249–58.
11. Von Dach E, Albrich WC, Brunel AS, Prendki V, Cuvelier C, Flury D, et al. Effect of C-reactive protein-guided antibiotic treatment duration, 7-day treatment, or 14-day treatment on 30-day clinical failure rate in patients with uncomplicated Gram-negative bacteremia: a randomized clinical trial. JAMA. 2020;323:2160–9.
12. Trenholme GM, Kaplan RL, Karakusis PH, Stine T, Fuhrer J, Landau W, et al. Clinical impact of rapid identification and susceptibility testing of bacterial blood culture isolates. J Clin Microbiol. 1989;27:1342–5.
13. Van Belkum A, Bachmann TT, Ludke G, Lisby JG, Kahlmeter G, Mohess A, et al. Developmental roadmap for antimicrobial susceptibility testing systems. Nat Rev Microbiol. 2019;17:51–62.
14. Gajic I, Kabic J, Kekic D, Jovicevic M, Milenkovic M, Mitic CD, et al. Antimicrobial susceptibility testing: a comprehensive review of currently used methods. Antibiotics (Basel). 2022;11:427.
15. Seng P, Drancourt M, Gouriet F, La Scola B, Fournier PE, Rolain JM, et al. Ongoing revolution in bacteriology: routine identification of bacteria by matrix-assisted laser desorption ionization time-of-flight mass spectrometry. Clin Infect Dis. 2009;49:543–51.
16. Carbonnelle E, Grohs P, Jacquier H, Day N, Tenza S, Dewailly A, et al. Robustness of two MALDI-TOF mass spectrometry systems for bacterial identification. J Microbiol Methods. 2012;89:133–6.
17. Jang KS, Kim YH. Rapid and robust MALDI-TOF MS techniques for microbial identification: a brief overview of their diverse applications. J Microbiol. 2018;56:209–16.
18. Mackay IM. Real-time PCR in the microbiology laboratory. Clin Microbiol Infect. 2004;10:190–212.
19. Wojno KJ, Baunoch D, Luke N, Opel M, Korman H, Kelly C, et al. Multiplex PCR based urinary tract infection (UTI) analysis compared to traditional urine culture in identifying significant pathogens in symptomatic patients. Urology. 2020;136:119–26.
20. Routh AF. Drying of thin colloidal films. Rep Prog Phys. 2013;76:46603.
21. Parsa M, Harmand S, Sefiane K. Mechanisms of pattern formation from dried sessile drops. Adv Colloid Interface Sci. 2018;254:22–47.
22. Zang D, Tarafdar S, Tarasevich YY, Dutta Choudhury M, Dutta T. Evaporation of a droplet: from physics to applications. Phys Rep. 2019;804:1–56.
23. Chen R, Zhang L, Zang D, Shen W. Blood drop patterns: formation and applications. Adv Colloid Interface Sci. 2016;231:1–14.
24. Pal A, Gope A, Sengupta A. Drying of bio-colloidal sessile droplets: advances, applications, and perspectives. Adv Colloid Interface Sci. 2023;314:102870.
25. Sefiane K, Duursma G, Arif A. Patterns from dried drops as a characterisation and healthcare diagnosis technique, potential and challenges: a review. Adv Colloid Interface Sci. 2021;298:102546.
26. Vaez S, Dadfar B, Koenig M, Franzreb M, Lahann J. Deep learning-based classification of histone-DNA interactions using drying droplet patterns. Small Sci. 2024;4:2400252.
27. Dadfar B, Vaez S, Haret C, Koenig M, Mohammadi Hafshejani T, Franzreb M, et al. Deep-learning-assisted affinity classification for humoral immunoprotein complexes. Small Struct. 2024;5:2400204.
28. Jeihanipour A, Lahann J. Deep-learning-assisted stratification of amyloid beta mutants using drying droplet patterns. Adv Mater. 2022;34:2110404.
29. Thokchom AK, Swaminathan R, Singh A. Fluid flow and particle dynamics inside an evaporating droplet containing live bacteria displaying chemotaxis. Langmuir. 2014;30:12144–53.
30. Majee S, Chowdhury AR, Pinto R, Chattopadhyay A, Agharkar AN, Chakravortty D, et al. Spatiotemporal evaporating droplet dynamics on fomites enhances long term bacterial pathogenesis. Commun Biol. 2021;4:1173.
31. Rasheed A, Hegde O, Chatterjee R, Sampathirao SR, Chakravortty D, Basu S. Physics of self-assembly and morpho-topological changes of *Klebsiella pneumoniae* in desiccating sessile droplets. J Colloid Interface Sci. 2023;629:620–31.

32. Sempels W, De Dier R, Mizuno H, Hofkens J, Vermant J. Auto-production of biosurfactants reverses the coffee ring effect in a bacterial system. Nat Commun. 2013;4:1757.
33. Ma X, Liu Z, Zeng W, Lin T, Tian X, Cheng X. Crack patterns of drying dense bacterial suspensions. Soft Matter. 2022;18:5239–48.
34. Han GR, Goncharov A, Eryilmaz M, Ye S, Palanisamy B, Ghosh R, et al. Machine learning in point-of-care testing: innovations, challenges, and opportunities. Nat Commun. 2025;16:3165.
35. Riti J, Sutra G, Naas T, Volland H, Simon S, Perez-Toralla K. Combining deep learning and droplet microfluidics for rapid and label-free antimicrobial susceptibility testing of colistin. Biosens Bioelectron. 2024;257:116301.
36. Moen E, Bannon D, Kudo T, Graf W, Covert M, Van Valen D. Deep learning for cellular image analysis. Nat Methods. 2019;16:1233–46.
37. Deegan RD, Bakajin O, Dupont TF, Huber G, Nagel SR, Witten TA. Capillary flow as the cause of ring stains from dried liquid drops. Nature. 1997;389:827.
38. Kasyap TV, Koch DL, Wu M. Bacterial collective motion near the contact line of an evaporating sessile drop. Phys Fluids. 2014. https://doi.org/10.1128/aac.00355-19.
39. Chen G, Mohamed GJ. Complex protein patterns formation via salt-induced self-assembly and droplet evaporation. Eur Phys J E. 2010;33:19–26.
40. Lee J, Jha K, Harper CE, Zhang W, Ramsukh M, Bouklas N, et al. Determining the Young 's modulus of the bacterial cell envelope. ACS Biomater Sci Eng. 2024;10:2956–66.
41. Vadillo-Rodriguez V, Pedraz P, Flors C. How much force is needed to kill a single bacterium? Small. 2025;21:e2407990.
42. Tadesse LF, Ho CS, Chen DH, Arami H, Banaei N, Gambhir SS, et al. Plasmonic and electrostatic interactions enable uniformly enhanced liquid bacterial surface-enhanced raman scattering (SERS). Nano Lett. 2020;20:7655–61.
43. Osborn MJ. Structure and biosynthesis of the bacterial cell wall. Annu Rev Biochem. 1969;38:501–38.
44. Talo M, Yildirim O, Baloglu UB, Aydin G, Acharya UR. Convolutional neural networks for multi-class brain disease detection using MRI images. Comput Med Imaging Graph. 2019;78:101673.
45. Dawood H, Nawaz M, Ilyas MU, Nazir T, Javed A. Attention-guided CenterNet deep learning approach for lung cancer detection. Comput Biol Med. 2025;186:109613.
46. Sauget M, Valot B, Bertrand X, Hocquet D. Can MALDI-TOF mass spectrometry reasonably type bacteria? Trends Microbiol. 2017;25:447–55.
47. Clark AE, Kaleta EJ, Arora A, Wolk DM. Matrix-assisted laser desorption ionization-time of flight mass spectrometry: a fundamental shift in the routine practice of clinical microbiology. Clin Microbiol Rev. 2013;26:547–603.
48. Boolchandani M, D'Souza AW, Dantas G. Sequencing-based methods and resources to study antimicrobial resistance. Nat Rev Genet. 2019;20:356–70.
49. Lee S, Park JS, Hong JH, Woo H, Lee CH, Yoon JH, et al. Artificial intelligence in bacterial diagnostics and antimicrobial susceptibility testing: current advances and future prospects. Biosens Bioelectron. 2025;280:117399.
50. Kim MJ, Park J, Kang M, Jeong U, Jeong D, Kang NG, et al. Bacteria detection and species identification at the single-cell level using super-resolution fluorescence imaging and AI analysis. Biosens Bioelectron. 2023;240:115603.
51. Kim S, Lee MH, Wiwasuku T, Day AS, Youngme S, Hwang DS, et al. Human sensor-inspired supervised machine learning of smartphone-based paper microfluidic analysis for bacterial species classification. Biosens Bioelectron. 2021;188:113335.
52. Ozel DB, Hadadi N, Babu AF, van der Meer JR. Rapid detection of microbiota cell type diversity using machine-learned classification of flow cytometry data. Commun Biol. 2020;3:379.
53. Rho E, Kim M, Cho SH, Choi B, Park H, Jang H, et al. Separation-free bacterial identification in arbitrary media via deep neural network-based SERS analysis. Biosens Bioelectron. 2022;202:113991.
54. Tseng YM, Chen KL, Chao PH, Han YY, Huang NT. Deep learning-assisted surface-enhanced raman scattering for rapid bacterial identification. ACS Appl Mater Interfaces. 2023;15:26398–406.
55. Zou D, Deng K, Zeng Z, Gong W, Guo B, Liu Y, et al. A bioinspired lipopeptide surmounts therapeutic dilemmas of Gram-positive and Gram-negative polymicrobial infection. J Am Chem Soc. 2025;147:36339–51.
56. Liu PY, Wu HC, Li YL, Cheng HW, Liou CH, Chen FJ, et al. Comprehensive pathogen identification and antimicrobial resistance prediction from positive blood cultures using nanopore sequencing technology. Genome Med. 2024;16:141.

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